PHY202 Practice Exam 7 on chapter(s): 27, 28

Name:____

These questions assess your understanding of the material from Chapter(s) 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. You shine a beam of white light through a diffraction grating with 5.7×10^{3} lines per centimeter to a screen $4.0\,\mathrm{m}$ away. Find the angle for first-order diffraction of $680\,\mathrm{nm}$ light.
(Chapter 27, Section (4) Multiple-Slit Diffraction - cp1e_ch27_ex3)

   ANSWER: ____________________________

2. Calculate the energy that is released upon the complete annihilation of a $17.\,\mathrm{g}$ object.
(Chapter 28, Section (6) Relativistic Energy - cp1e_ch28_ex6)

   ANSWER: ____________________________

3. Suppose a space probe is moving away from Earth at a speed of $1.5 \times 10^{8}\,\mathrm{m/s}$. It sends a radio-wave message back to Earth at a frequency of $1.2\,\mathrm{GHz}$. At what frequency is the message received on Earth?
(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5b)

   ANSWER: ____________________________

4. Suppose you pass light from a laser through two slits separated by $0.024\,\mathrm{mm}$ and find that the third bright line on a screen is formed at an angle of 5.9° relative to the incident beam. What is the wavelength of the light?
(Chapter 27, Section (3) Young's Double-Slit Experiment - cp1e_ch27_ex1)

   ANSWER: ____________________________

5. Suppose a cosmic ray colliding with a nucleus in Earth's upper atmosphere produces a muon that has a velocity equal to $2.3 \times 10^{8}\,\mathrm{m/s}$. The muon then travels at constant velocity and lives for a duration of $0.23\,\mu\mathrm{s}$ in the muon's frame of reference. How long does the muon live as measured by an Earth-bound observer?
(Chapter 28, Section (2) Simultaneity and Time Dilation - cp1e_ch28_ex1)

   ANSWER: ____________________________

6. Suppose a galaxy is moving away from Earth at a speed of $2.7 \times 10^{8}\,\mathrm{m/s}$. It emits radio waves with a wavelength of $0.33\,\mathrm{m}$. What wavelength would we detect on Earth?
(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5)

   ANSWER: ____________________________

7. Suppose an astronaut is traveling at $1.5 \times 10^{8}\,\mathrm{m/s}$ from Earth to Alpha Centauri, which is a start system $8.2\,\mathrm{ly}$ away as measured by an Earth-bound observer. How far apart are Earth and Alpha Centauri as measured by the astronaut? Note: ly is lightyear, the distance light travels in one year.
(Chapter 28, Section (3) Length Contraction - cp1e_ch28_ex2)

   ANSWER: ____________________________

8. The primary mirror of an orbiting space telescope has a diameter of $0.50\ \text{m}$. One of the key advantages of operating in space is that it avoids atmospheric distortion, allowing the telescope to approach its diffraction-limited resolution. Assuming the telescope observes visible light with a wavelength of $500\ \text{nm}$, what is the minimum angular separation between two point sources that can be just resolved according to the Rayleigh criterion? Report your answer in arcseconds. Note: 1 degree is 60 arcminutes and 1 arcminute is 60 arcseconds.
(Chapter 27, Section (6) Limits of Resolution - cp1e_ch27_ex5)

ANSWER: ____________________________

9. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity $1.5\ \text{kW/m}^2$ to reduce the intensity to $0.73\ \text{kW/m}^2$?
(Chapter 27, Section (8) Polarization - cp1e_ch27_p87)

ANSWER: ____________________________

10. Visible light of wavelength $530\ \text{nm}$ falls on a single slit and produces its second diffraction minimum at an angle of 7.3° relative to the incident direction of the light. What is the width of the slit?
(Chapter 27, Section (5) Single-Slit Diffraction - cp1e_ch27_ex4)

ANSWER: ____________________________

11. The headlights of a car have a wavelength of about $600\ \text{nm}$ and are 0.53-m apart. Assuming the eye's pupil diameter is $0.42\ \text{cm}$, what is the maximum distance at which the two headlights can be resolved? Use the diffraction limit (Rayleigh criterion) for your calculation.
(Chapter 27, Section (6) Limits of Resolution - cp1e_ch27_p66)

ANSWER: ____________________________

12. Calculate the minimum thickness of an oil slick on water that appears blue when illuminated by white light perpendicular to its surface. Take the blue wavelength to be $460\ \text{nm}$, $n_{\text{oil}} = 1.48$ and $n_{\text{water}} = 1.34$.
(Chapter 27, Section (7) Thin-Film Interference - cp1e_ch27_p72)

ANSWER: ____________________________

cp1e: College Physics (1st Edition) by OpenStax
PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway
tkd: I did not refer to anything while writing this question